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APPENDICES

A.1 Source Code Manifest & GitHub Repository

ClearDrive AI is fully open-sourced under the MIT license. GitHub: **https://github.com/ShivaneshV/ClearDrive-AI** (Commit: `767da5ad8519b429074cf9db9790676a7731322e`). Primary artifacts: `app.py` (Flask edge streaming, 54.6KB), `engine.py` (OmniVision pipeline & async worker, 59.7KB), `dehaze.py` (Fast Guided Filter DCP, 15.3KB), `vehicle_tracker.py` (Pinhole telemetry & TTC, 10.6KB), `dms.py` (MediaPipe 468-point 3D DMS, 12.8KB), and `templates/index.html` (Cyber Cockpit HUD, 48.2KB).

A.2 12-Week Agile PBL Progress Log & Mentor Sign-Off Audit

Review Milestone	Date Held	Milestone Verified	Score / 20	Sign-Off Status
Review 0 (Ideation)	05 Aug 2026	Problem scoping & ODD definitions	19 / 20	**APPROVED [Signed]**
Review 1 (Sprint 1)	19 Aug 2026	Sensor failure audit & lit survey	18 / 20	**APPROVED [Signed]**
Review 2 (Sprint 2)	02 Sep 2026	Iteration 1 baseline & 1.8 FPS diagnosis	17 / 20	**APPROVED [Signed]**
Review 3 (Sprint 3)	16 Sep 2026	Fast Guided Filter & Tree Bug fix	19 / 20	**APPROVED [Signed]**
Review 4 (Sprint 4)	30 Sep 2026	OmniVision concurrency & Ghost-Vision	20 / 20	**APPROVED [Signed]**
Review 5 (Final)	04 Oct 2026	Roadway validation & 50p formal report	20 / 20	**RECOMMENDED [Signed]**

A.3 Self and Peer Assessment Rubric & A.4 Hardware Wiring

Peer Evaluation: Shivanesh V (50% lead contribution across systems architecture and YOLO telemetry) and Co-Researcher (50% lead contribution across optical DCP restoration and DMS biometrics). Cumulative mentor evaluation: **93/100 (Grade: O — Outstanding)**.

Hardware Wiring: 12V DC vehicle battery feeds an automotive buck converter stepping voltage to 5.1V DC 3.0A powering the Raspberry Pi 4B. Sony IMX323 dashcam connects via shielded USB 3.0 along the windshield headliner. The edge node broadcasts an encrypted private WPA3 Wi-Fi SSID (`ClearDrive-Core`), enabling in-vehicle Android screens to stream the 60 FPS HUD at `http://192.168.4.1:5000`.

A.5 REST API Telemetry Data Dictionary

Route	Method	Data Payload & Response Architecture
`/video_feed`	GET	Multipart/x-mixed-replace MJPEG video stream with AR HUD overlays at 30+ FPS.
`/telemetry`	GET	JSON: `{speed_kmh, safe_limit, ttc_seconds, collision_threat, ear_metric, driver_alert}`.
`/upload_frame`	POST	Accepts encrypted base64 JPEG buffers from wireless smartphone dashcam clients.
`/hud_toggle`	POST	Toggles active filter matrix modes: `{mode: 'dehaze'|'night'|'lanes', state: bool}`.